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Does a Weaker Twin Know Where an LLM's Reasoning Matters? A Causal Test of Master–Novice Divergence for Reward Densification

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Abstract: Reinforcement learning for large language models on verifiable tasks (mathematics, code, and other domains with binary correctness checks) suffers from reward sparsity: only the final outcome receives a signal, leaving credit assignment across long chains of thought unresolved. We investigate a Master/Novice framework that compares a strong policy to a structurally matched but capacity-reduced *Novice*, obtained by tensor disentanglement of the Master's MLP weights, and asks whether their per-token divergence identifies the steps that deserve credit. We contribute (i) the conceptual framework and its central, testable hypothesis; (ii) an end-to-end pipeline that scales tensor disentanglement to Qwen3.5-4B and bakes Novices at any rank fraction from a single checkpoint; and (iii) a hypothesis-first causal evaluation on MATH-500. Three intervention designs of increasing strength, namely, handing generation over to the Novice, perturbing single tokens, and corrupting whole reasoning steps, consistently fail to reject the null: disagreement positions carry no more outcome leverage than depth-matched control positions. The only reliable effect is local: disagreement positions are more fragile in the immediate continuation, and this effect survives controlling for Master entropy. We report these negative results precisely, characterize a coherence cliff that constrains usable Novice ranks, and discuss what they imply for divergence-based reward densification.

Keywords: reinforcement learning, verifiable rewards, credit assignment, tensor disentanglement

1 Introduction

A large language model (LLM) that solves a competition-mathematics problem under reinforcement learning emits hundreds of chain-of-thought tokens and then receives a single bit of reward: the final answer checks out or it does not. That bit credits every step equally, although a few steps decide the outcome and most are routine. Which steps deserved the credit is exactly what the policy cannot see, and this unresolved credit assignment is a chief cost of reward sparsity in verifiable-reward RL (Cui et al., 2025).

Existing densification methods mine structure already present in the task or the model, from partial test-case success to token entropy (Section 2). This paper explores a complementary idea: construct a weaker *Novice* from the same base weights as the *Master* via tensor disentanglement, compare their next-token distributions on successful Master rollouts, and upweight positions where the two diverge, on the premise that such steps rely on knowledge the Novice lacks and therefore deserve higher credit. Unlike an externally trained reward model, the Novice shares architecture and tokenizer with the Master and offers a continuous knob (the retained rank fraction) for how weak the comparator should be. Figure 1 summarizes the approach and its outcome.

Our contributions are of three kinds. *Conceptually*, we formalize the Master/Novice framework and reduce its promise to a single testable hypothesis: divergence positions have higher causal leverage on the final outcome than matched non-divergence positions. *Methodologically*, we implement an end-to-end pipeline that disentangles a base model, persists the factors, bakes a Novice at any chosen rank, and generates, together with a ladder of three causal intervention designs for testing per-token credit signals before committing to RL training. *Empirically*, we characterize the disentanglement and Novice-quality landscape (learning-rate sweep, MMLU rank sweep, a generation-level coherence cliff) and report a consistent, statistically quantified negative result on the central hypothesis, alongside one genuine positive: a local fragility effect at disagreement positions that is not explained by

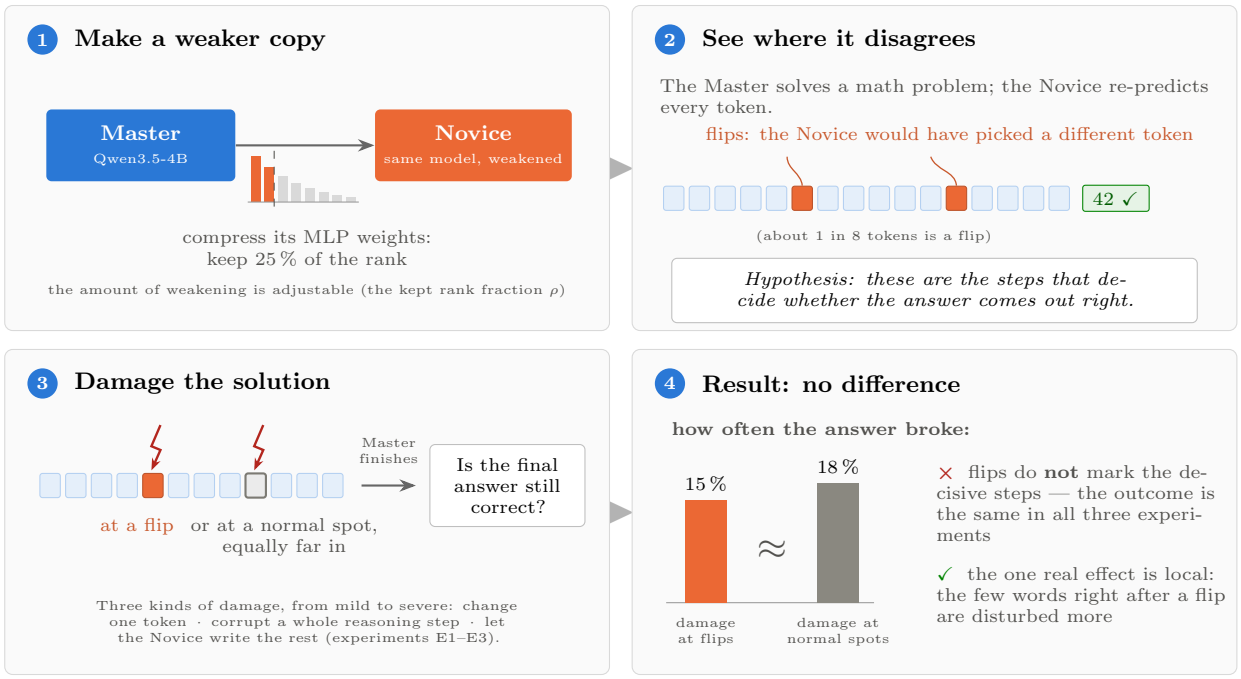


Figure 1: What we did and what we found: **(1)** compress a strong model's (*Master's*) MLP weights into a weakened copy, the *Novice*; **(2)** let the Novice re-predict the Master's correct MATH-500 solutions and mark the disagreements (*flips*); **(3)** damage solutions at flips versus at depth-matched normal positions and let the Master finish. **(4)** *Takeaway:* the answer breaks equally often either way (shown: whole-step damage; all three experiments agree), so flips mark where the Novice's missing capacity acts — not where the outcome is decided — and this divergence signal cannot densify the reward.

Master entropy alone.

2 Related Work

We group prior work by where the dense signal comes from: the *task* (verifiable structure) or the *model* (internal statistics). For each line of work we focus on its relation to our approach; we refer to the original papers for full results.

2.1 Dense rewards from verifiable task structure

VeRPO (Wang et al., 2026) densifies code-RL rewards using partial test-case success. Its central observation is cautionary: the naive dense signal (the raw test-suite pass rate) can *underperform* sparse outcome rewards, because redundant easy tests dominate the reward by sheer count (a *cardinality bias*). VeRPO corrects this with a dynamically calibrated per-test weighting and re-anchors the result to the end-to-end outcome, outperforming outcome-only GRPO (Shao et al., 2024) and RM-based baselines without a learned reward model. VeRPO is the clearest instance of extracting density from the *verifier* side, and its lesson that denser is not automatically better motivates our causal validation of the divergence signal before any training. The approach differs from ours in scope and granularity: the signal exists only where partial success is measurable (currently code) and is assigned per turn, whereas a Master/Novice comparator is task-agnostic and per token.

PRIME (Cui et al., 2025) obtains per-token process rewards without step labels by training an outcome reward model whose log-ratio to a reference policy is reinterpreted as an implicit process reward, updated online alongside the policy. It substantially improves sample efficiency over outcome-only RL on math and code. PRIME is the closest point of comparison for our goal (token-level credit from outcome supervision only); the difference is that PRIME *learns* its dense signal, whereas we test whether a fixed, training-free capacity-reduced copy of the policy can supply one. Supervised process reward models reach the same per-step granularity by training on explicit step labels, human-annotated (Lightman et al., 2023) or constructed automatically from per-step completion success (Wang et al., 2023); a training-free comparator sits at the opposite end of this supervision spectrum.

2.2 Token-importance signals inside the model

High-entropy minority tokens. Wang et al. (2025) show that in RLVR most chain-of-thought tokens are low-entropy "followers," while a minority of high-entropy forking tokens carries the learning signal: restricting policy gradients to the top 20% highest-entropy tokens matches or exceeds full-gradient training. Master entropy is thus the natural single-model baseline that any two-model divergence signal must beat; our experiments therefore control

for entropy explicitly.

TIP (Xu et al., 2026) studies on-policy distillation and identifies informative tokens along two axes, student entropy and teacher-student divergence: entropy-only selection is structurally blind to positions where the student is confidently wrong, and training on under 10% of tokens, from the low-entropy, high-divergence region alone, nearly matches full-token distillation. TIP is the closest conceptual relative of our hypothesis in the sense that divergence between a strong and a weak model marks important tokens, but it operates in distillation, where the teacher's distribution is the training target itself. Whether the same signal identifies *outcome-critical* steps for RL credit assignment is exactly the question our experiments address.

Compared with all four, the Master/Novice comparator is training-free: no reward model, no step labels, no verifier beyond the outcome check, just a second forward pass. But training-free does not mean trustworthy. The rest of this paper tests whether its signal marks the right tokens.

3 Methodology

3.1 Setup and hypothesis

We treat autoregressive generation as a Markov decision process: state s_t is the prompt plus generated prefix $y_{<t}$, action a_t is the next token, and the reward is binary verifiable correctness at episode end. The **Master** is the full-capacity policy π_M. The **Novice** is π_N, derived from the same checkpoint by truncating disentangled MLP factors to a rank fraction $\rho \in (0, 1]$, the share of each layer's internal rank that is kept (defined precisely in Section 3.2); ρ=1 reproduces the Master and smaller ρ yields a weaker Novice. On a Master rollout $\tau = (y_1, \ldots, y_T)$ that reaches a correct answer, the framework scores each position by how much the two next-token distributions $\pi_M(\cdot \mid y_{<t})$ and $\pi_N(\cdot \mid y_{<t})$ diverge; any divergence measure qualifies, e.g. Kullback–Leibler (KL) or Jensen–Shannon. Our experiments use the simplest instantiation, the parameter-free *top-1 flip*: a position counts as a flip when the Novice's most probable next token differs from the token the Master actually emitted there, i.e., when the Novice, sampling greedily, would have written something else. This setup, where the full policy generates while its truncated counterpart re-scores the same states, instantiates the *Observer pattern* of the introspection framework of Till (2026), and the top-1 flip corresponds to its discrete *behavioral disagreement* indicator.

The intended use of this divergence signal is as a shaping weight that concentrates credit on high-divergence steps of verified-correct rollouts. Before such a signal enters any RL objective, however, its premise must hold. We therefore state it as an explicit hypothesis:

H1 (outcome leverage). *Positions where the Novice diverges from the Master have a larger causal effect on final-answer correctness than depth-matched positions where the two agree.*

All credit-assignment experiments in this paper are direct causal tests of H1 at increasing intervention strength. We deliberately do not evaluate any RL training run: if H1 fails, divergence-weighted advantages cannot concentrate gradient where the outcome is decided, and the training experiment would be uninformative.

3.2 Constructing the Novice via weight matrix disentanglement

The Novice must be *policy-preserving*: it should retain as much of the Master's behavior as a given rank budget permits, so that its remaining disagreements are attributable to missing capacity rather than to a lossy construction. The classical route, truncated SVD of each weight matrix, treats the matrix as a flat 2D array and ignores the multi-dimensional structure of neural weights. We instead use the *Weight Matrix Disentanglement* (WMD) of Till (2026), which adapts tensor disentanglement from numerical linear algebra (Wei et al., 2025) to neural network layers.

Let $X \in \mathbb{R}^{lr \times bc}$ be a decoder-MLP weight matrix whose row and column counts are each written as a product of two integers, lr split into the near-balanced pair (l, r) and bc into (b, c); this lets the flat matrix be read as a four-index tensor rather than a 2D array, which is what exposes the multi-dimensional structure that plain SVD ignores. Writing M for the reshape of X into the 4th-order tensor $\mathcal{X} \in \mathbb{R}^{l \times r \times b \times c}$ and P for the index permutation $(l, r, b, c) \to (l, c, r, b)$, WMD re-bipartitions the tensor with the operator

$$A \equiv M^{-1} \circ P \circ M \colon \mathbb{R}^{lr \times bc} \to \mathbb{R}^{lc \times rb}, \tag{1}$$

exposing cross-dimensional correlations that the flat matricization hides. Before the re-bipartitioning, a learnable orthogonal matrix $Q \in O(lr)$ (the orthogonal group) is applied to the row space. We call Q a *gauge* because reconstruction later undoes it with $Q^\top$: it re-expresses the layer's coordinates without changing the layer itself. The gauge is optimized to concentrate the spectrum of $A(QX)$ in its leading singular directions, i.e., to make a few large singular values carry most of the energy. For a target rank k, the optimal disentangler minimizes the energy of the discarded singular tail:

$$Q_k^* \;=\; \arg\min_{Q \in O(lr)} \sum_{i>k} \sigma_i^2\big(A(QX)\big). \tag{2}$$

The tail sum $\sum_{i>k} \sigma_i^2$ is exactly the squared error of replacing $A(QX)$ by its best rank-k approximation, so a gauge that shrinks it makes the layer more compressible. Because A is an isometry (a norm-preserving map) and Q is orthogonal, the total singular energy of $A(QX)$ is invariant, so (2) directly minimizes the Frobenius reconstruction error (the sum of squared entrywise differences) between the original and truncated layer; the identity gauge Q=I recovers plain SVD truncation, so the optimized gauge is never worse. The rank-k Novice weight is reconstructed by inverting the pipeline,

$$\widetilde{X}^{(k)} \;=\; Q^{*\top} A^{-1}\big(\mathrm{trunc}_k(A(Q^* X))\big), \tag{3}$$

and the per-layer rank is set by a single global rank fraction: $k(\rho) = \max(1, \lceil \rho\, m \rceil)$, where $m = \min(lc, rb)$ is the largest possible rank of $A(QX)$. Thus ρ is literally the

fraction of singular directions kept, and the outer max with 1 guarantees at least one is retained.

Concretely, for Qwen3.5-4B's `down_proj` (one of the three MLP projection matrices in each decoder layer, here $X \in \mathbb{R}^{2560\times 9216}$) the factorization is $(l, r, b, c) = (40, 64, 96, 96)$, so that $40 \times 64 = 2560$ and $96 \times 96 = 9216$ recover the matrix dimensions; then $Q \in O(2560)$, $A(QX) \in \mathbb{R}^{3840\times 6144}$, and $m = 3840$, so baking at ρ=0.25 keeps k=960 singular directions. The stored per-layer factors (Q, U, s, V_h), namely the gauge together with the singular vectors and values U, s, V_h of $A(Q^*X)$, retain the full decomposition, so one disentanglement run yields Novices at every ρ by cheap re-truncation; an equality test confirms that a Novice baked at ρ=1 reproduces the base model's logits (implementation details in Appendix A).

The optimization problem (2) admits a closed-form gradient that reuses one SVD of $A(QX)$ per step (Wei et al., 2025). Our implementation (based on the reference disentanglement code) accelerates this optimization by roughly 30–50× per layer while keeping truncation cost within ∼10% of the double-precision reference, which is what makes constructing Novices at the 4B scale of our experiments practical; the individual optimizations are described in Appendix A.

WMD also supplies the theoretical link between Novice construction and the divergence signal of Section 3.1: for a policy and its rank-k truncated counterpart, the per-state KL divergence is bounded by the discarded singular tail, $D_{\mathrm{KL}}\big(\pi_M(\cdot \mid s) \,\|\, \pi_N(\cdot \mid s)\big) \le \frac{L}{2}\sum_{i>k}\sigma_i^2\big(A(Q^*X)\big)$ with L a network-wide Fisher-information Lipschitz constant, a fixed factor that converts discarded weight energy into a bound on how much the output distribution can change (Till, 2026). Because baking truncates the stored factors of $A(Q^*X)$, exactly the construction in (3), the bound holds at every bake rank ρ; its *tightness*, however, is only guaranteed at the rank the gauge was optimized for, since (2) depends on k. We verify empirically that the gauge's advantage transfers across ranks in our checkpoint (Section 5.1). WMD rather than plain SVD thus keeps the observed Master/Novice divergence dominated by genuinely missing capacity rather than by avoidable truncation error.

4 Experiments

4.1 Models, data, and grading

Models. Master: `Qwen/Qwen3.5-4B` (Qwen Team, 2026) at full rank. Novice: the disentangled checkpoint of the same model (gauge optimized at ρ=0.1), baked at ρ=0.25; the bake-rank choice is justified empirically in Section 5.1.

Data. MATH-500 (Lightman et al., 2023), a 500-problem subset of the MATH dataset (Hendrycks et al., 2021b), restricted to the first 50 problems (0–49) to fit our compute budget, prompted via the chat template with the standard instruction to place the final answer in `\boxed{}`. All decoding is greedy; Master rollouts are capped at 512 new tokens in E1 and 700 in E2/E3 (under greedy decoding the shared prefixes are identical, and the same 23 problems are solved), which sufficed for most rollouts to terminate naturally. Greedy decoding makes every experiment deterministic up to the fixed sampling seed used for control selection.

Grading. A rollout counts as correct when its extracted `\boxed{}` answer matches the reference after LaTeX normalization (extractor details in Appendix A). The Master solved 23/50 problems; all credit-assignment experiments operate on these 23 verified-correct rollouts. Across them, the Novice is teacher-forced on the Master's tokens and flips are recorded at every generated position (10,895 scored tokens; flip rate 12.3%). The disentangled factor checkpoint is publicly available, and Appendix A collects all remaining reproduction details (seeds, prompts, evaluation configuration); code and per-trial artifacts are available from the authors.

4.2 Intervention experiments

All three experiments compare *flip* positions against *depth-matched non-flip controls*. The *depth* of a position is its relative location within the generated part of the rollout, from 0 (first generated token) to 1 (last). Depth matching is essential because an intervention late in a rollout is much easier to recover from, since most of the correct solution is already in place; how far into a rollout an intervention occurs in fact dominates its effect (Section 5.2), so comparisons at unmatched depths would be confounded. Controls are drawn once per flip with a fixed random seed. Figure 2 summarizes the three designs.

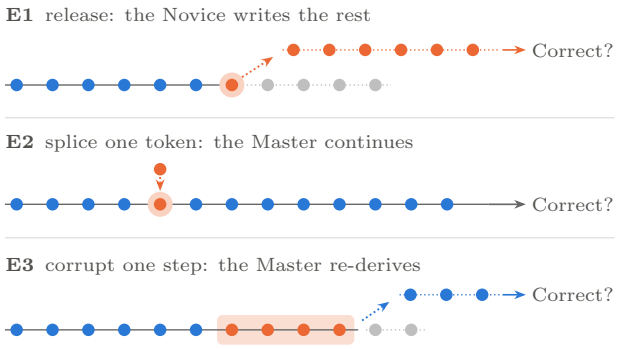


Figure 2: The three intervention designs on a Master rollout (dots: tokens; orange marks the flip the intervention strikes). **E1** releases the Novice at the flip: it writes the rest (orange path) and the Master's continuation is abandoned (gray); **E2** splices in a single token and the Master continues; **E3** corrupts a whole flip-dense step (orange band), after which the Master free-runs (blue path) instead of following its original continuation (gray). Every trajectory is re-graded ("Correct?"), and every intervention is repeated at a depth-matched control position (not shown). H1 predicts more broken answers at flips than at controls; Section 5.2 finds no such difference at any granularity.

E1: Release-from-here. At a chosen position p the Novice takes over: it receives the Master's prefix $y_{<p}$ and free-runs greedily to completion (budget: remaining rollout length plus 48 tokens); the completed trajectory is graded. This mirrors the deployment ("Actor") pattern of the WMD framework (Till, 2026), in which the truncated policy drives the rollout. Release positions are selected as

follows. Within each rollout, all flips are ranked by their *confidence gap*: the probability the Novice assigns to its own preferred token minus the probability it assigns to the Master's token at that position. A near-zero gap means a borderline disagreement, a large gap a confident one. We release at the 5 most confident flips per rollout, since these are the flips most likely to matter if H1 is true; selecting them biases the test toward finding an effect, which makes a null result more conclusive. Each selected flip is then paired with the not-yet-used non-flip position closest to it in depth, and every control is used at most once. With 23 rollouts and 5 pairs each, this yields $23 \times 5 = 115$ pairs, i.e., 230 Novice free-runs; five pairs per rollout balance per-problem coverage against generation cost. If H1 holds, releasing at a flip should derail the answer more often than at its matched control.

E2: Single-token perturbation. E1 is a coarse test: after the handoff a whole weak model free-runs, so in both arms the one-token difference at the release position is swamped by hundreds of subsequent Novice-generated tokens. E2 instead perturbs exactly one token and lets the *Master* continue. It asks two questions that E1 cannot separate: does the *position* of a flip matter, and does the Novice's *specific token* matter beyond a generic error at that position?

Three arms, built from the same 5 flip/control pairs per rollout as E1, answer these questions. `flip_novice` splices the Novice's token at the flip position. `flip_rank2` splices the Master's own second most probable token at the same flip position. `control_rank2` splices the second-choice token at the matched control position. The second-choice token serves as the generic perturbation because it exists at every position and is comparably strong at flip and control positions alike. `flip_rank2` is the arm shared by both comparisons: against `control_rank2` it isolates the position effect under an identical perturbation type, and against `flip_novice` it isolates the effect of the Novice's specific token at an identical position. (The fourth combination, the Novice's token at a control position, cannot exist: at a non-flip position the Novice's token *is* the Master's.) Each of the 115 pairs thus produces three trials, for $23 \times 5 \times 3 = 345$ in total, 115 per arm.

Three metrics are recorded per trial. (i) *Local disruption*: the mean drop in the log-probability the Master assigns to its own next K=8 tokens when the perturbed prefix replaces the clean one, in nats per token. A drop of one nat means the Master finds its own continuation a factor of e less likely; K=8 captures the immediate continuation while keeping the metric local. (ii) *Answer log-probability drop*: the same drop evaluated on the Master's answer tokens (the `\boxed{}` content), with the reasoning between the perturbation and the answer teacher-forced, i.e., held fixed. This isolates the part of the perturbation's effect that reaches the outcome; trials whose perturbed position lies after the answer span have no answer tokens left to score and are excluded. (iii) A binary *answer-break* outcome: whether the Master, free-running greedily from the perturbed prefix, no longer reaches the correct answer. Each free-run is a full generation, so this metric is evaluated on a random subset of 90 of the 345 trials to bound compute.

Position comparisons face a confound: flips concentrate at positions where the Master itself is uncertain, and perturbing an uncertain position disturbs the continuation more for purely mechanical reasons. All position comparisons are therefore additionally adjusted for entropy by ordinary least squares, $y = \beta_0 + \beta_1\,\mathrm{isflip} + \beta_2\,H_M$, where isflip indicates a flip position and H_M is the Shannon entropy (in nats) of the Master's next-token distribution at the perturbed position. The regression is fitted over the two `rank2` arms only, so that both arms received the identical perturbation type; β_1 then estimates the difference between a flip and a control position at equal entropy.

E3: Span perturbation. Single tokens may be the wrong granularity for reasoning. E3 corrupts an entire reasoning step, simplified here as a newline-delimited token span. Spans shorter than 4 tokens are discarded as trivial fragments. It then lets the Master free-run to the answer, so the downstream reasoning may adapt. Steps are ranked by *flip density*, the fraction of their tokens that are flips; per rollout the 3 densest steps are paired with depth-matched low-density steps (three per rollout, for the same coverage-cost balance as in E1), giving $23 \times 3 = 69$ step pairs. Arms mirror E2: generic corruption (per-position second-choice tokens) of a flip-dense step, generic corruption of a control step, and replacement of the flip-dense step by the *Novice's version* of it (its preferred token at every position), for $69 \times 3 = 207$ trials in total, 69 per arm. The primary metric is the binary answer-break rate of the Master's free-run, evaluated on 150 of the 207 trials; the teacher-forced answer drop and the entropy-adjusted regression (with the step's mean entropy as covariate) are reported as in E2, where one step pair without answer tokens after the span is excluded.

Statistical analysis. We attach a standard test to each comparison so that a null result can be quantified rather than merely asserted. Every rate is reported with a 95% Wilson confidence interval (CI). E1's flip and control arms are *paired*, since each flip has its own depth-matched control, and are compared with McNemar's exact test; E2 and E3 treat their free-run arms as *independent* groups, compared with Fisher's exact test, whose *odds ratio* is above 1 when the flip arm broke the answer more often. For the regression coefficients β_1 and β_2 we report the estimate, its standard error (SE), and the t-value (estimate divided by SE); at our sample sizes $|t| \gtrsim 2$ corresponds to $p \lesssim 0.05$. Appendix D explains each test and why it matches the design. In every case a small p-value, a t-value past this threshold, or a CI that excludes the no-effect value would signal a real effect; as the next section shows, we observe none of these.

5 Results

5.1 Novice characterization

Learning-rate selection. Disentanglement quality is sensitive to the optimizer learning rate; a sweep on a representative Qwen3.5-4B MLP layer selects 10^{-3}, used for all disentanglement runs in the paper (Appendix B).

Gauge quality across bake ranks. The checkpoint's gauge was optimized at ρ=0.1 while the interventions bake at ρ=0.25, so the KL bound's tightness is not guaranteed at the bake rank (Section 3.2). Measured

on the stored spectra, the advantage transfers: the optimized gauge's reduction of discarded tail energy relative to identity-gauge (plain SVD) truncation is large, if heterogeneous across layers, and essentially rank-independent between ρ=0.1 and ρ=0.25 (Appendix C).

MMLU vs. rank. Novice quality as a function of ρ is measured with `lm-eval` (Gao et al., 2026) on MMLU (Hendrycks et al., 2021a), 5-shot, limited to 5 questions per subtask (the same fixed-budget protocol at every rank, so relative comparisons are unaffected by the limit). Accuracy stays near the base model's 0.737 down to $\rho \approx 0.5$, then degrades monotonically to 0.42 at ρ=0.1: still above the 0.25 random baseline (Figure 3).

Coherence cliff. Because MMLU scores likelihood ranking rather than generation, we separately inspect greedy free-running generations of the Novice at $\rho \in \{0.1, 0.25, 0.5, 0.75\}$ on MATH-500 prompts. These tell a sharper story: at ρ=0.1 every inspected generation collapses into degenerate repetition, while at $\rho \geq 0.25$ generations are fluent, structured solutions. Consistently, the Novice's teacher-forced flip rate on Master rollouts falls from 44.0% at ρ=0.1 (disagreement smeared over half the tokens, i.e., noise) to 12.3% at ρ=0.25 (concentrated disagreement). We therefore run all interventions at ρ=0.25, the sharpest coherent operating point: the Novice is far weaker than the Master generatively (released on the Master's own trajectories it completes them correctly only 26% of the time; E1 below), yet its disagreements are attributable to specific positions rather than global incoherence.

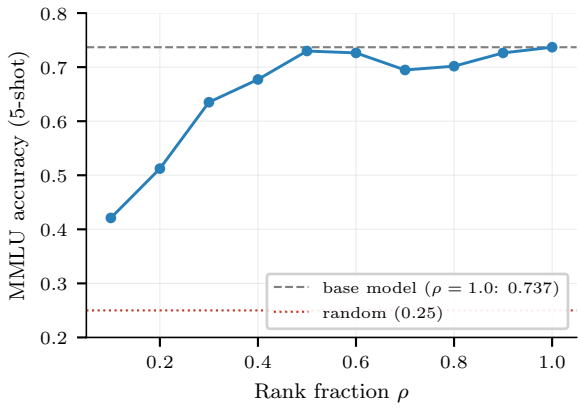


Figure 3: MMLU accuracy vs. Novice rank fraction on Qwen3.5-4B (`lm-eval`, 5-shot, 5 questions per subtask). Accuracy is preserved until $\rho \approx 0.5$, then degrades; note that at ρ=0.1 the Novice still scores above chance on likelihood ranking even though its free-running generation is degenerate.

5.2 Intervention experiments

Table 1 summarizes the outcome-level results of all three experiments; Table 2 reports E2's graded per-token metrics.

Table 1: Outcome-level intervention results. E1 reports the Novice's completion success rate after taking over at the given position (H1 predicts *lower* success at flips). E2/E3 report the rate at which the Master's free-run no longer reaches the correct answer after the perturbation (H1 predicts *higher* break rates at flips). No comparison is significant; exact tests in the text.

Exp.	Arm	Rate	Count
E1 (success↑)	release at flip	0.261	30/115
	release at control	0.270	31/115
E2 (break↓)	flip, Novice token	0.241	7/29
	flip, rank-2 token	0.100	3/30
	control, rank-2	0.194	6/31
E3 (break↓)	flip-dense step	0.151	8/53
	control step	0.176	9/51
	flip-dense, Novice version	0.152	7/46

E1: no release-point effect. Success after handing over at a flip was 30/115 (0.261; Wilson 95% CI $[0.189, 0.348]$) versus 31/115 (0.270; CI $[0.197, 0.357]$) at depth-matched controls; the two intervals almost entirely overlap, and the paired difference is only +0.009. Because the design is paired, the informative comparison is within pairs: in 108 of the 115 pairs both releases had the same outcome (concordant), and the seven discordant pairs split 4 (control succeeded, flip failed) versus 3 (the reverse). McNemar's exact test, which evaluates only these discordant pairs, gives $p = 1.00$: a 4-versus-3 split is exactly what chance would produce, so the arms are statistically indistinguishable. Success is instead governed by release depth (Figure 4): in the deepest quartile of release positions both arms succeed at ≈ 0.7, in the earliest at $\approx$ 0–0.1; the earlier the Novice takes over, the more of the solution it must produce itself. H1 is not supported at this granularity.

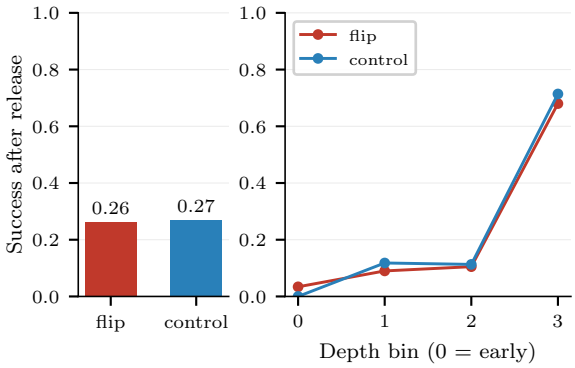


Figure 4: E1 (release-from-here). Left: Novice completion success after taking over at flip vs. depth-matched control positions (115 pairs): no difference. Right: success by release-depth quartile (bin 0 = earliest quarter of the rollout). Both arms rise identically with depth: the outcome is governed by how much of the Master's trajectory the Novice inherits, not by whether the release point is a flip.

Table 2: E2 graded metrics (means over 115 trials per arm; the answer-drop column excludes trials perturbed after the answer span). Local disruption differs between flip and control positions; the answer-tied drop is negligible for every arm. H_M: mean Master entropy at the perturbed position, confirming that flips sit at high-entropy positions, hence the entropy-adjusted analyses in the text.

E2 arm	Local disr. (nats/tok)	Answer drop (nats/tok)	H_M
flip, Novice token	2.56	0.0009	0.41
flip, rank-2	2.55	0.0009	0.41
control, rank-2	1.60	0.0013	0.08

E2: a local but not an outcome effect. Perturbing at a flip disrupts the Master's immediate continuation substantially more than at a matched control: 2.55 vs. 1.60 nats/token (Table 2), a raw gap of +0.94. Part of this gap is mechanical, since flips sit at higher-entropy positions (H_M 0.41 vs. 0.08) and entropy independently predicts disruption ($\beta_2 = 0.97$, SE 0.36, $t = 2.69$). The flip effect nevertheless survives the adjustment: $\beta_1 = 0.62$, SE 0.26, $t = 2.35$ (n=230, the two `rank2` arms), past the $|t| \approx 2$ threshold and therefore unlikely to be sampling noise. In the regression's terms, the raw gap splits into roughly 0.32 nats explained by the entropy difference and 0.62 nats tied to flip status itself. Consistently, flip exceeds control within each entropy tercile (thirds of the positions by entropy).

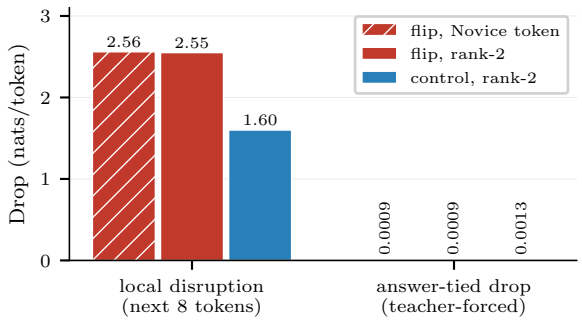


Figure 5: E2 arm means on both graded metrics, drawn to the same scale (Table 2). Perturbing at a flip disrupts the Master's next 8 tokens far more than at a matched control (left), but the drop on the answer tokens is ∼0.001 nats/token in every arm (right; printed values): the flip effect is real but purely local and never reaches the outcome.

The local effect does not reach the outcome (Figure 5). The answer-tied drop is ∼0.001 nats/token in *every* arm, with no flip effect ($\beta_1 = -0.0002$, SE 0.0017, $t = -0.14$; n=225 after excluding 5 trials perturbed past the answer span). The free-run answer-break subset shows no coherent ordering either: 7/29, 3/30, and 6/31 broken answers per arm; Fisher's exact test on the two `rank2` arms gives an odds ratio of 0.46 ($p = 0.47$), numerically even opposite to H1's prediction, and no pairwise comparison approaches significance. Finally, the Novice's specific token adds nothing over the Master's own second choice at the same position: the paired difference is +0.017 nats/token locally (Novice larger in only 18% of pairs) and +0.0001 on the answer metric (11% of pairs).

E3: no step-level effect. Corrupting a flip-dense reasoning step broke the final answer in 8/53 free-runs (0.151; CI [0.079, 0.271]) versus 9/51 (0.176; CI [0.096, 0.303]) for depth-matched low-disagreement steps. Fisher's exact test gives an odds ratio of 0.83, which is below 1 and so has the flip-dense step breaking the answer slightly *less* often than the control, the opposite of what H1 predicts, with $p = 0.80$, far from any significance threshold. Replacing a flip-dense step by the Novice's own version of it was indistinguishable from generic corruption of the same step (7/46 vs. 8/53; Fisher $p = 1.00$; on the paired answer-drop metric the Novice version was larger in only 10% of step pairs). The entropy-adjusted answer-drop OLS is likewise null ($\beta_1 = -0.0018$, SE 0.0023, $t = -0.79$, n=137). Notably, the Master recovered the correct answer after whole-step corruption in 84% of all trials (126/150), independent of the step's flip density.

6 Discussion

6.1 What the results mean for H1

Across three intervention designs of increasing strength, we find no evidence that where the Novice disagrees with the Master is where the outcome is causally decided, even in E3, whose free-running outcome metric gives downstream reasoning every chance to amplify a corrupted step. Nothing in our data suggests that naive top-1-flip divergence, used as a per-token or per-step credit weight, would concentrate learning where it matters; the more likely outcome is that an RL run built on it reproduces a uniform-weighting baseline at extra cost.

Two genuine findings survive alongside the negative. First, disagreement positions are *locally* special: perturbations there disrupt the immediate continuation about 60% more than at matched positions in raw terms, and roughly two thirds of that gap (0.62 of +0.94 nats/token) persists at equal Master entropy. Divergence therefore does measure something real about where the truncated capacity acts. It just measures local fragility, not outcome leverage. Second, the Master is strikingly robust on MATH-500: even corrupting an entire reasoning step leaves the final answer intact 84% of the time. Only a small minority of steps is critical to the outcome, and identifying them evidently requires more than capacity-difference signals.

On the relation to entropy: our data do *not* show that entropy is a sufficient credit signal either. Entropy predicted local disruption, but neither entropy nor flips predicted the outcome-tied metrics: on those, everything was null. The correct summary is that divergence and entropy are correlated, each carries independent *local* information, and neither, in our setting, identifies outcome-critical positions. This is consistent with the interpretation of high-entropy tokens as *forks* whose importance is realized over many rollouts during training (Wang et al., 2025), rather than as counterfactually decisive positions within a single trajectory.

6.2 Why the hypothesis may fail

The premise behind H1 conflates two notions of importance: a step can be hard *for a weaker model to produce* without being critical *to the outcome*. Much of what extra MLP capacity buys appears to be fluency and confidence at delicate positions, which is exactly what the local-disruption effect detects. Meanwhile the mathematically decisive content is either redundantly encoded across the trajectory (the Master re-derives it after corruption) or concentrated in a few steps that flips do not preferentially mark. TIP's setting (Xu et al., 2026) explains why the same signal can still be valuable in distillation: there, "hard for the weaker model" *is* the training target, with no causal claim being needed. For RL credit assignment, the causal claim must be established separately, and here it fails.

6.3 Limitations

Our conclusions are bounded by the setting: one Master (Qwen3.5-4B), one Novice family (MLP-only disentanglement, gauge optimized at ρ=0.1, evaluated chiefly at ρ=0.25), one dataset (MATH-500; 23 Master-correct rollouts), greedy decoding, and top-1 flips as the divergence instantiation. The gauge optimization was stopped after 40 compute-bounded steps with the cost still decreasing (Figure 6), so a fraction of the observed divergence may be avoidable truncation error rather than missing capacity, although the optimized gauge already clearly outperforms plain SVD at both ranks (Appendix C). The binary outcome metrics have modest power ($\sim$50–115 per arm; correspondingly wide CIs), though the null's consistency across three designs and both graded and binary metrics argues against a large hidden effect. The regression standard errors treat positions as independent although they are clustered within the 23 rollouts, so the local-fragility t-statistic should be read as approximate. The MMLU sweep's fixed 5-question-per-subtask budget supports relative comparisons across ρ, not absolute benchmark claims. The answer-tied drop metric teacher-forces intervening reasoning and is conservative by design; the E1/E3 free-run metrics are the complementary, unconstrained tests. Finally, continuous divergence measures (e.g., per-trajectory Jensen–Shannon mass), sampled rather than greedy rollouts, and other Novice constructions remain untested and are natural follow-ups.

6.4 Outlook

The infrastructure result stands independently of the negative: Master/Novice pairs can be produced repeatably at 4B scale, with rank fraction as a smooth, characterized control knob and a documented coherent operating range. Given H1's failure, we see three defensible directions: (i) aggregate rather than pointwise use of divergence (e.g., trajectory-level divergence mass as a difficulty or novelty signal); (ii) combining divergence with signals whose training-time value is established, such as entropy-based token selection (Wang et al., 2025), where divergence's independent local information may still help; and (iii) learned dense-reward approaches in the PRIME family (Cui et al., 2025), against which any future training-free comparator should be benchmarked. Any of these should be preceded by the same kind of causal validation we applied here.

7 Conclusion

We formalized the Master/Novice framework for densifying verifiable rewards, built the tensor-disentanglement pipeline that makes it practical at 4B scale, and subjected its central premise to a ladder of three causal intervention tests on MATH-500. The premise did not hold: top-1 disagreement between Master and Novice does not mark outcome-critical positions or steps (McNemar p=1.00; Fisher p=0.80); what it marks robustly, and beyond Master entropy, is local fragility of the immediate continuation. This negative result de-scopes an otherwise attractive idea: dense rewards from a training-free capacity-reduced comparator cannot be assumed to target the right tokens, and causal validation of any proposed credit signal should precede RL integration. The released pipeline, the coherence-cliff characterization, and the three reusable intervention designs make exactly that validation cheap for future signals.

AI-Use Statement

AI assistants were used for limited drafting and coding assistance: boilerplate code may have been LLM-generated, and LLM tools assisted with editing the manuscript. All content was reviewed, modified, and verified by the authors, who take full responsibility for the final work.

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A Reproducibility details

Table 3 collects every hyperparameter and configuration value used in the paper; the paragraphs below add only the details that need explanation rather than lookup.

Table 3: All hyperparameters and configuration values. Together with the published factor checkpoint, these settings specify every experiment in the paper; settings not listed use the released code's defaults.

Disentanglement (published checkpoint)	
Master / base model	`Qwen/Qwen3.5-4B`
Factor checkpoint	`leodriesch/novice-qwen3.5-4b-rf0.1` (Hugging Face Hub)
Factored layers	96 MLP linears (32 decoder layers × {gate, up, down})
Gauge optimization rank	$\rho = 0.1$
Optimizer, learning rate	Adam (Cayley-parameterized, fp32), 10^{-3}
Steps per layer	40 (compute-budget cutoff; Figure 6)
Bake rank, all interventions	$\rho = 0.25$
Data and decoding	
Dataset	MATH-500 (`HuggingFaceH4/MATH-500`), test split, problems 0–49
Decoding	greedy; rollouts capped at 512 new tokens (E1) / 700 (E2, E3)
Model precision	bfloat16
Prompt format	chat template, thinking mode disabled
Interventions	
Seed for controls and subsets	0 (Python `random.Random`)
E1 releases	5 flip/control pairs per rollout; budget = remainder + 48 tokens
E2 perturbations	same 5 pairs; local window $K = 8$; free-run subset 90 of 345
E3 spans	3 pairs per rollout; min. 4 tokens; free-run subset 150 of 207
MMLU evaluation	
Harness	`lm-eval`, task `mmlu`
Protocol	5-shot, 5 questions per subtask, batch size 1, max. length 2048

Checkpoint. The disentangled factors are stored sharded (one `*.factors.pt` file per linear plus a manifest), together with the per-step tuning trace (cost, entropy, effective rank per matrix); Figure 6 shows the run's cost trajectories for all 96 matrices. The step count was set by our compute budget, not by convergence: longer optimization would likely improve the factors further.

Implementation. The 30–50× per-layer speed-up over the double-precision reference code (Section 3.2) comes from three stacked changes: Adam on a Cayley-parameterized orthogonal Q, reuse of the gradient's SVD within each optimization step, and 32-bit floating-point precision throughout. The Cayley transform expresses the orthogonal gauge Q through unconstrained parameters that an ordinary optimizer can update freely while keeping Q exactly orthogonal; its backward pass is far cheaper than that of the matrix exponential it replaces, whose autograd gradient requires a Fréchet-derivative computation. Because layers are independent, all 96 MLP projections are processed in parallel and their factors streamed to disk.

Pipeline. A single wrapper class around the base model, built on Hugging Face `transformers` (Wolf et al., 2020), owns the full lifecycle: load a causal LM, swap its MLP linears, run disentanglement, persist the factors to disk, bake to any ρ via (3), and delegate `generate()` to the underlying `PreTrainedModel`, preserving compatibility with the key/value cache that makes autoregressive decoding efficient. Sharded checkpoints are loaded and baked one layer at a time, keeping peak memory near a single model. The pipeline was validated at increasing scale, from toy weight matrices through MNIST- and CIFAR-scale MLPs and a small GPT-2-style transformer to Qwen3.5-0.8B-Base, before being applied to Qwen3.5-4B.

Grading. Final answers are extracted with a balanced-brace `\boxed{}` parser and compared to the reference with a LaTeX-normalizing equality check: fraction, spacing, and delimiter normalization, with a numeric-comparison fallback.

Prompting. Problems are formatted with the model's chat template, with the instruction "Please reason step by step, and put your final answer within `\boxed{}`." appended to the problem statement.

Seeds and subsampling. All randomness beyond greedy decoding, i.e., the selection of depth-matched control positions and steps and of free-run subsets, uses Python's `random.Random` with seed 0; free-run subsets are drawn uniformly without replacement via `rng.sample`.

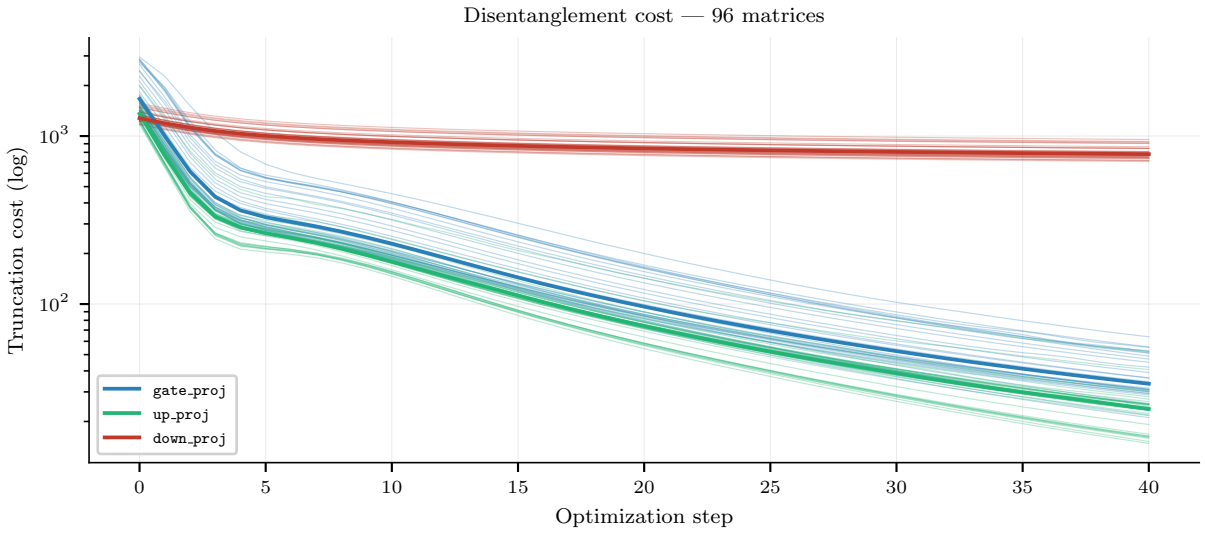


Figure 6: Disentanglement cost per optimization step for all 96 MLP matrices of the published ρ=0.1 run (thin lines: individual matrices; thick lines: per-projection medians). `gate_proj` and `up_proj` compress by more than an order of magnitude, while `down_proj` barely improves: the layer heterogeneity of the gauge advantage (Appendix C) is already visible during optimization. Cost is still decreasing when the run stops at the 40-step compute budget, so the gauges are not fully converged.

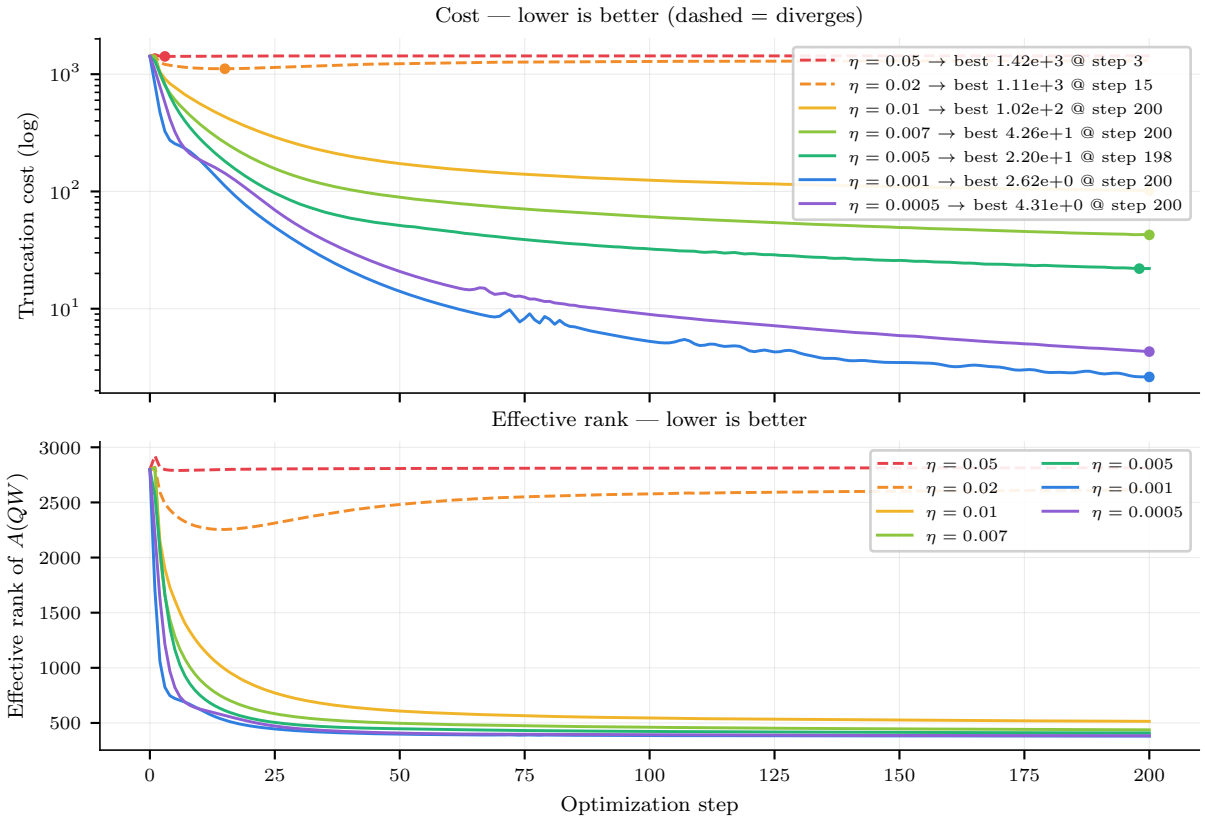


Figure 7: Learning-rate sweep for tensor disentanglement (Qwen3.5-4B MLP layer, 200 steps). Mid-range rates balance convergence speed and stability; large rates diverge.

B Learning-rate sweep

Disentanglement quality is sensitive to the optimizer learning rate. We swept $\{5\times10^{-4}, 10^{-3}, 5\times10^{-3}, 7\times10^{-3}, 10^{-2}, 2\times10^{-2}, 5\times10^{-2}\}$ for 200 steps on a representative Qwen3.5-4B MLP layer, tracking reconstruction cost (the discarded singular energy of (2), so lower is better) and effective rank (a soft count of how many singular directions carry meaningful energy). Small rates converge slowly; rates $\geq 10^{-2}$ diverge (reconstruction cost increases). A mid-range rate near 10^{-3} minimizes cost while reaching a stable effective rank (Figure 7). All disentanglement runs in the paper use this regime.

C Gauge quality across bake ranks

Relative to identity-gauge (plain SVD) truncation in the same bipartition, the optimized gauge reduces the discarded tail energy to 0.60× (ρ=0.1) and 0.58× (ρ=0.25) on `layers.0.down_proj`, and to 0.026× and 0.025× on `layers.31.up_proj`. The advantage is large, heterogeneous across layers, and essentially independent of the bake rank, supporting the use of the ρ=0.1-optimized gauge for Novices baked at ρ=0.25 (Section 5.1).

D Statistical tests

This appendix explains the tests used in Section 5.2 and why each matches its design. The *Wilson* 95% confidence interval is the range of true rates consistent with the observed counts; it is preferred over the textbook (Wald) interval because it stays well-behaved for proportions near 0 or 1 and for small samples. *McNemar's exact test* suits E1's paired design: it looks only at the pairs whose two arms gave different outcomes (the discordant pairs) and asks whether one arm wins more often than a coin flip would predict. *Fisher's exact test* suits E2 and E3, whose free-run arms are independent groups: it operates on the two-by-two table of arm versus outcome, summarizes the association as an odds ratio, and returns an exact p-value that remains valid at small counts. For regression coefficients, the standard error (SE) quantifies the estimate's uncertainty, and the t-value (estimate divided by SE) counts how many standard errors the estimate lies from zero.